

Lecture 3
Introduction to
Big Data Analytics
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Big Data Analytics and Business Intelligence (CSET 371)

Course Contents:

Module I:

8 lecture hours

Big Data Analytics: Data and Relations, Business Intelligence, Business intelligence vs business analytics, Big Data Predictive Analysis, Why what and how BI?, OLTP VS OLAP, Project life cycle, Methodology, Impact Analysis, Testing and release management, Ethics in Business Intelligence, Big Data Technology Component, Real Time Analysis of Big Data, Big Data Architecture, Cost Conscious Design, Storage Conscious Design.

Modulo II:

6 lecture hours

Big Data Warehouse, Functional vs Procedural programming models, Data placement strategies, CAP Theorem, Streaming, Stream Data Types, Stream Computing, Filtering Stream, Estimating Moments, NoSQL, Aggregate Data Models, Document Data Model, Schema Less Databases.

Module III:

6 lecture hours

Hadoop Ecosystem, Hadoop Overview, HDFS, Map-Reduce, Pig Overview, Pig Grunt Shell, Hive, HBase, Architecture of Hive and HBase.

Module IV:

8 lecture hours

HQL, Associations and Joins, Aggregate function, Polymorphic queries, Clauses, Subqueries, Spark, Core, Spark SQL, Spark RDD, Deployment and Integration, Spark GraphX and Graph Analytics, Research Tonics in Big Data Analytics

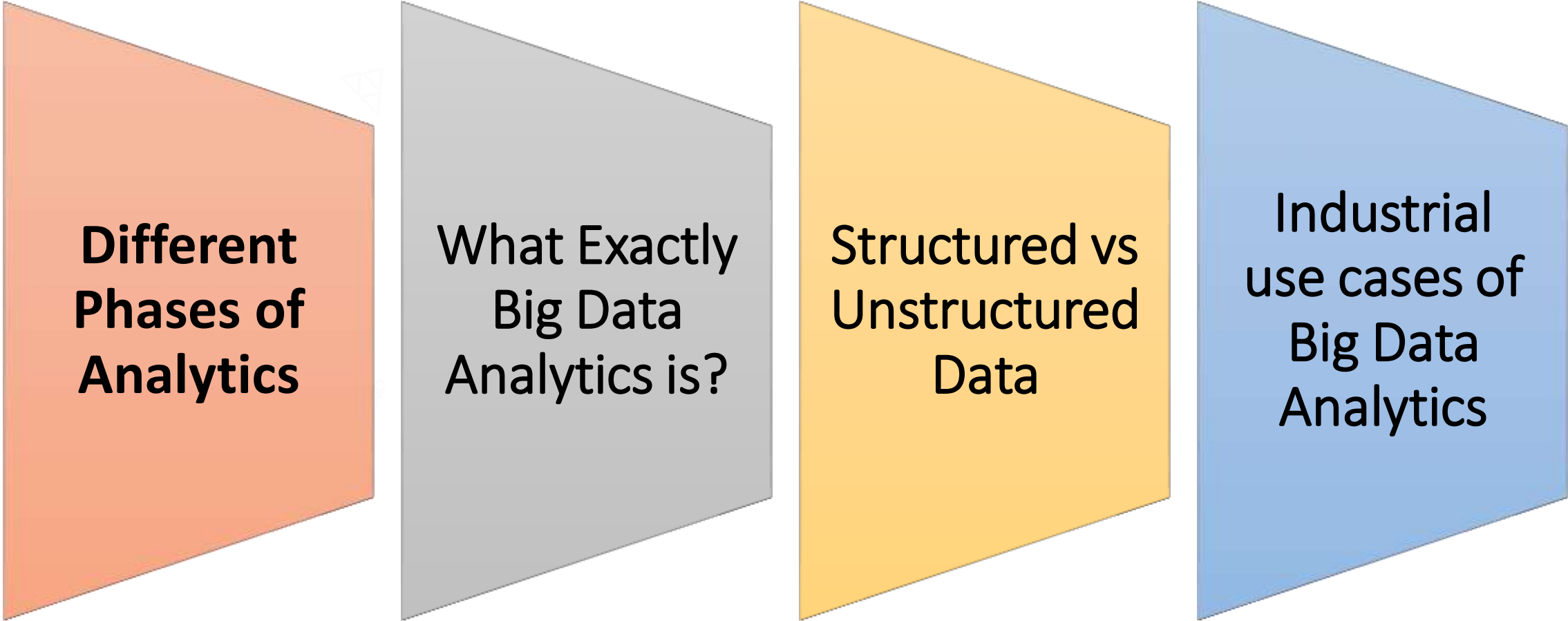
Text Books :

1. Ghavami Peter , *Big Data Analytics Methods* (2nd ed.), De Gruyter, 2020. ISBN 9781547417951.
2. Acharya and Seema, *Data Analytics using R* (1st ed.), New York: McGraw-Hill Education, 2018. ISBN 9352605241.

Reference Books :

1. Azevedo Ana and Manuel Filipe Santos, *Integration Challenges for Analytics, Business Intelligence, and Data Mining* (1st ed.), Engineering Science Reference, 2020. ISBN 9781799857832.

Learning Objectives



**Different
Phases of
Analytics**

**What Exactly
Big Data
Analytics is?**

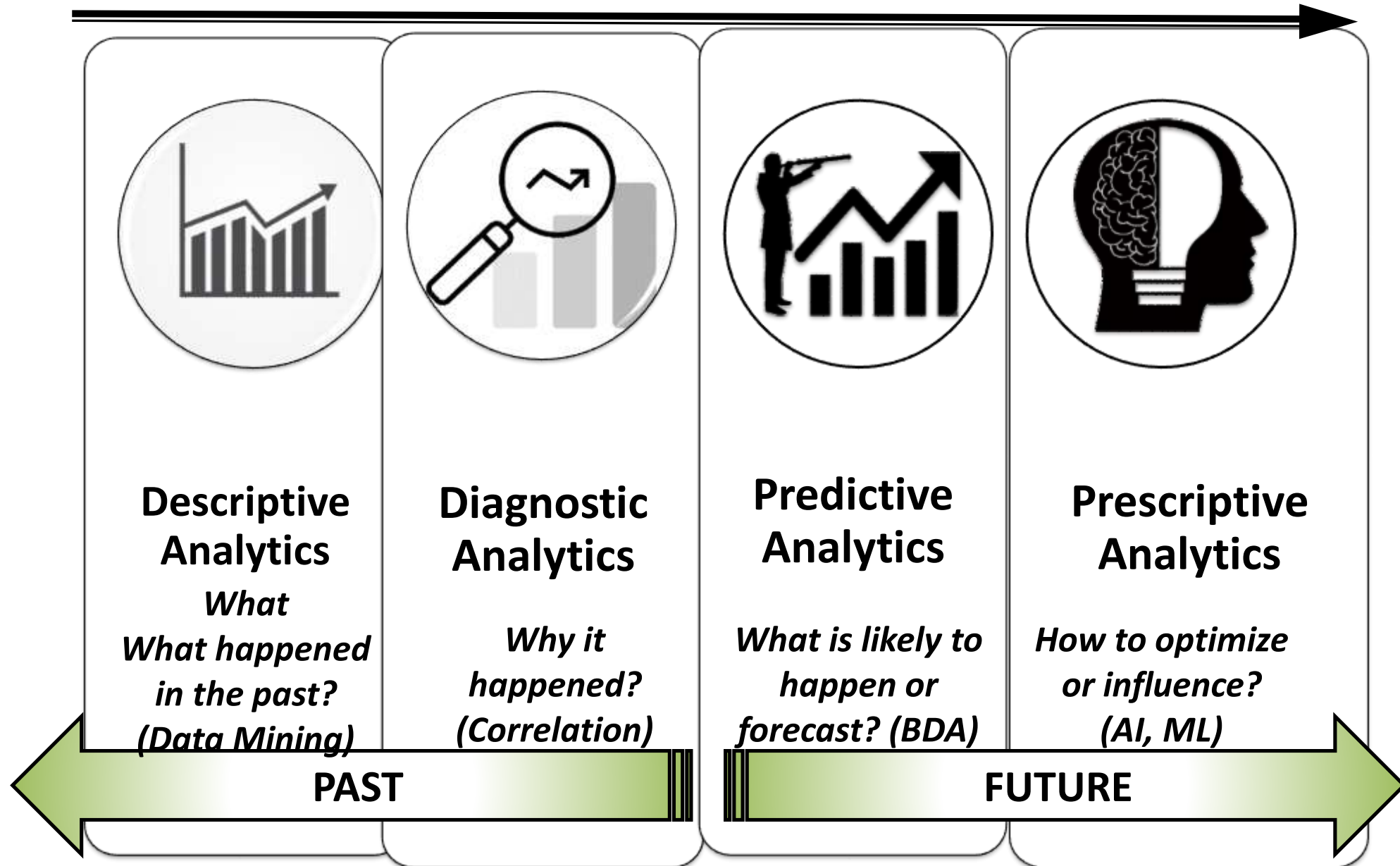
**Structured vs
Unstructured
Data**

**Industrial
use cases of
Big Data
Analytics**

Data Analysis

Data Analysis is collecting, cleaning, organizing, transforming, and modelling the data to extract useful information to support industrial decision-making.

Types of Data Analytics



Excercise

1. Business Intelligence (BI) is based on the type of analytics.?
2. Big Data Analytics focuses on what type of analytics?

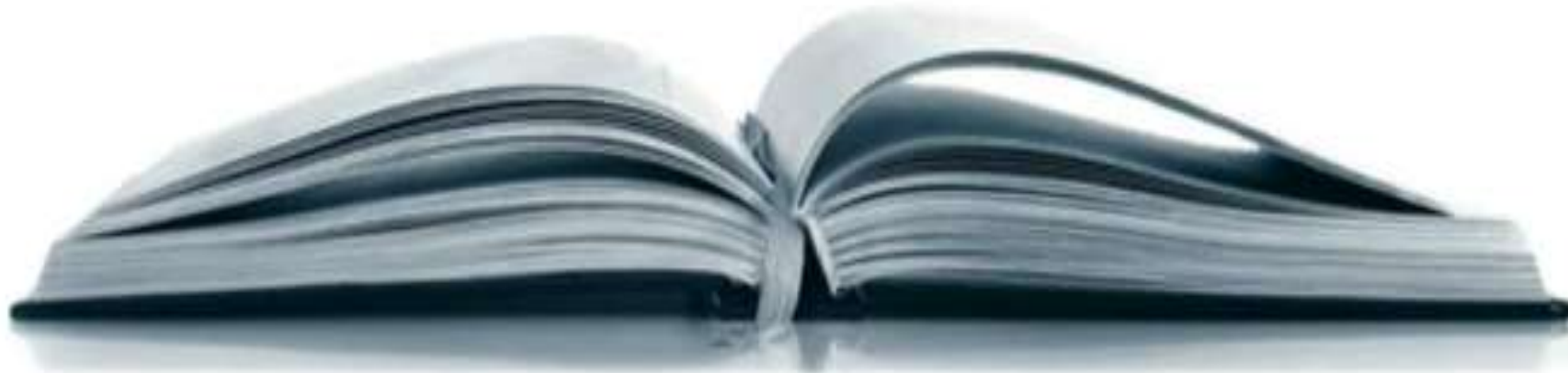
Sol.

Sol 1- Descriptive and diagnostic.

Sol 2- All- Big data analytics is not limited to one type but integrates descriptive, diagnostic, predictive, and prescriptive analytics to provide comprehensive insights and actionable recommendations.

What is Big Data?

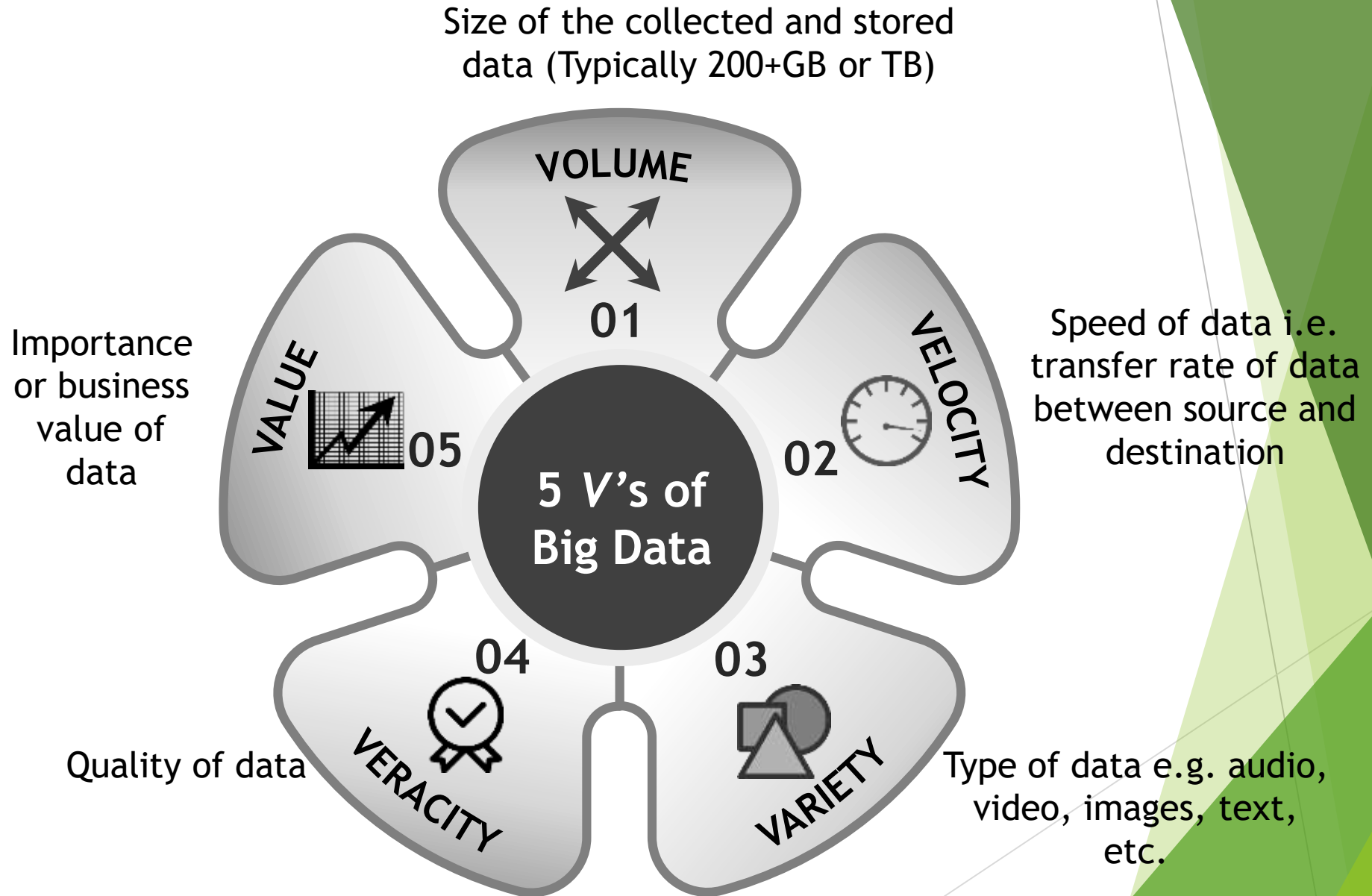
Big Data is an umbrella term for datasets that traditional technologies or tools can not practically handle due to their high volume, velocity and variety.



When does Data Becomes Big Data?

In order to say data is known as Big Data it has to satisfied 5 aspects: 5Vs

Big Data Characteristics



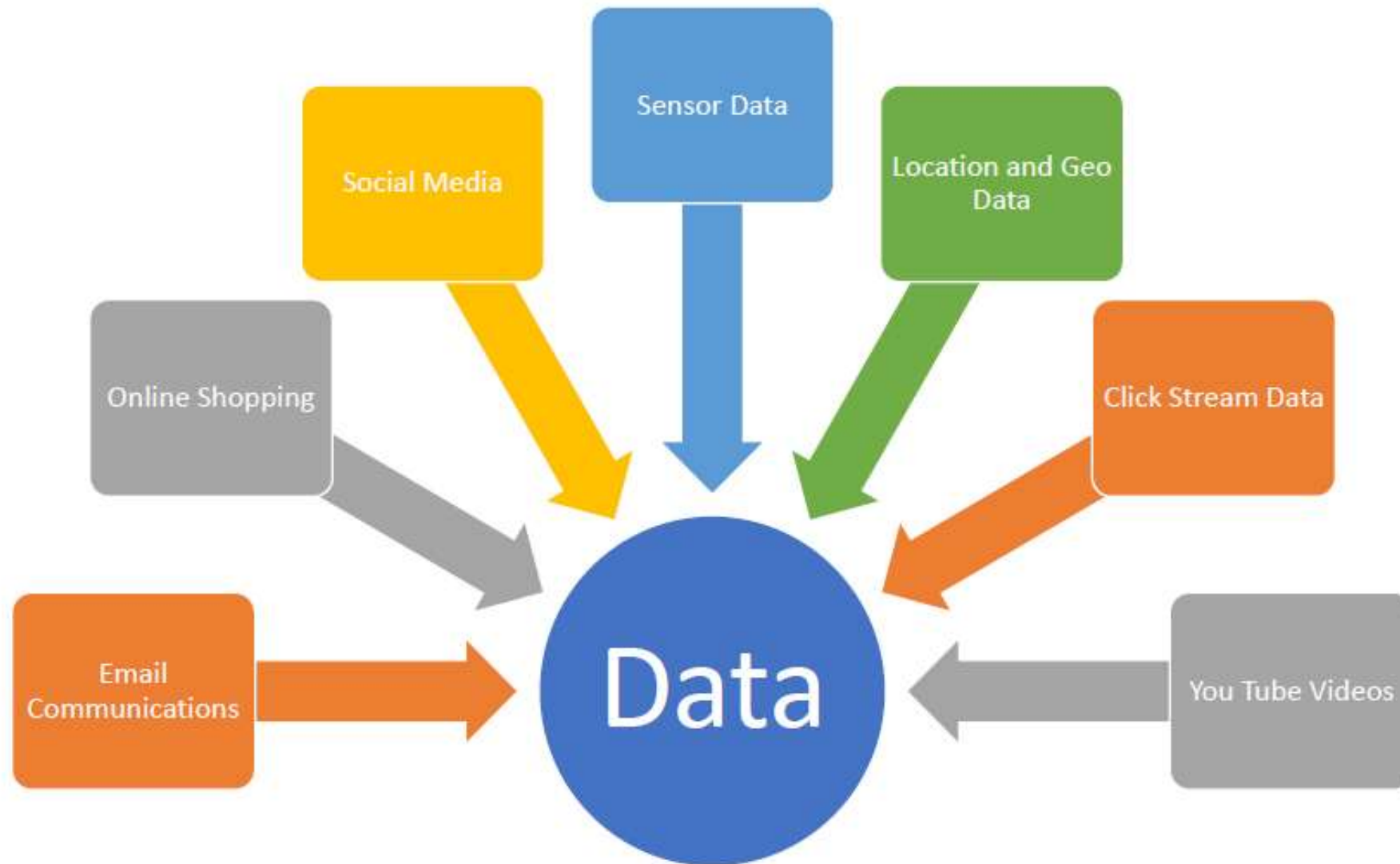
Approximate Data Size Thresholds for MySQL:

Hundreds of GBs to a Few TBs. As the database size grows beyond a few terabytes, MySQL may encounter performance issues, such as slow query responses, increased latency, and challenges in handling high volumes of concurrent transactions.

When to Consider Big Data Tools:

Data Size: When the data grows beyond a few terabytes and continues to scale rapidly, or when you're dealing with petabytes (PBs) of data, big data tools like Hadoop, Spark, or NoSQL databases (e.g., Cassandra, MongoDB) become more appropriate.

SOURCES OF BIG DATA



Types of Digital Data

Structured

- Data Organized in the form of rows and columns.
- Examples : RDBMS, Microsoft Excel

Semi structured

- Has some structure, but does not conform to any data model.
- Examples : All markup languages, HTML, XML etc

Unstructured

- Does not confirm to any data model.
- Examples : Media files, data from sensors and physical devices.

DATA SOURCES



DWH, DM,
OLTP, ODS



STRUCTURED



NoSQL



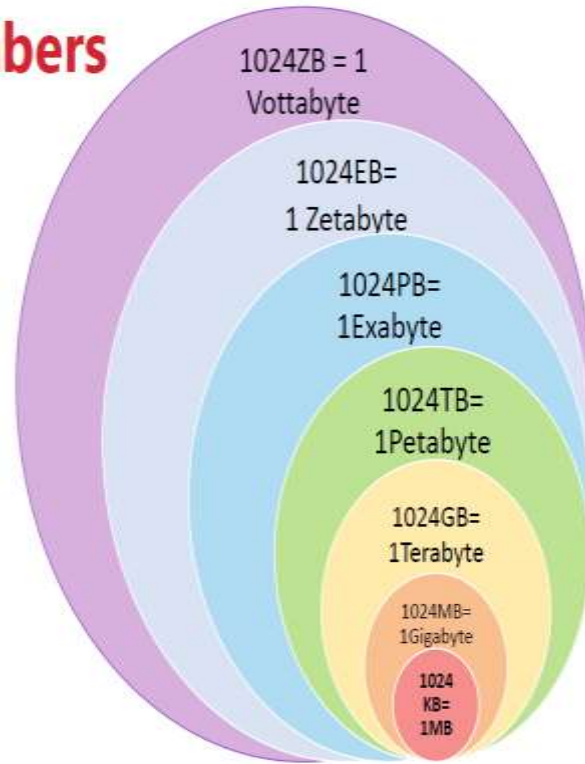
SEMI-STRUCTURED



UN-STRUCTURED

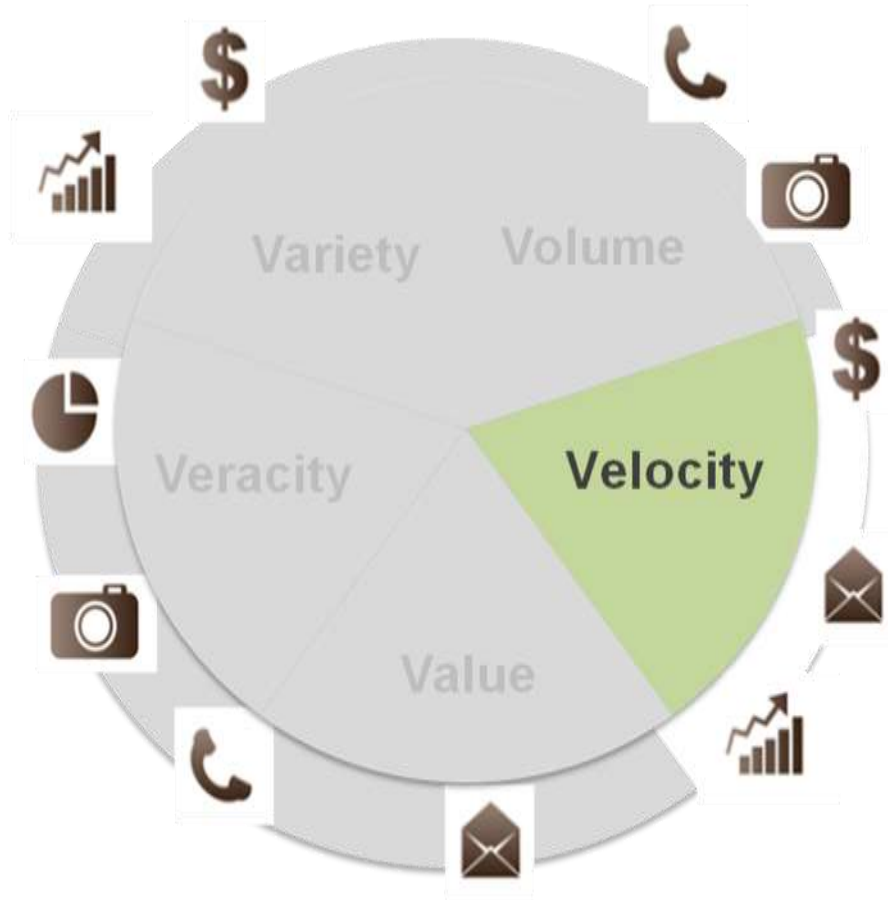
1. Volume

Big Data Numbers



Volume refers to the vast amounts of data generated every second. We are not talking Terabytes but Zettabytes or yottabyte. This makes most data sets too large to store and analyse using traditional database technology.

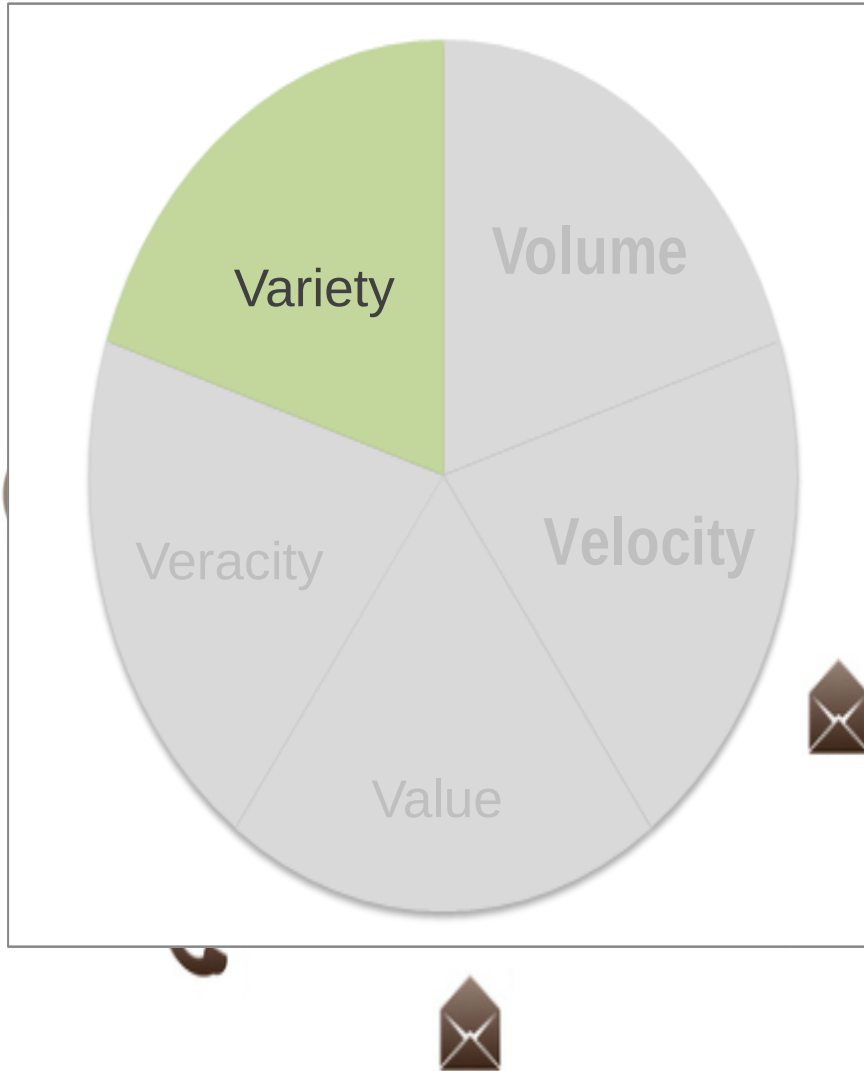
2. Velocity



Velocity refers to the speed at which new data is generated and the speed at which data moves around.

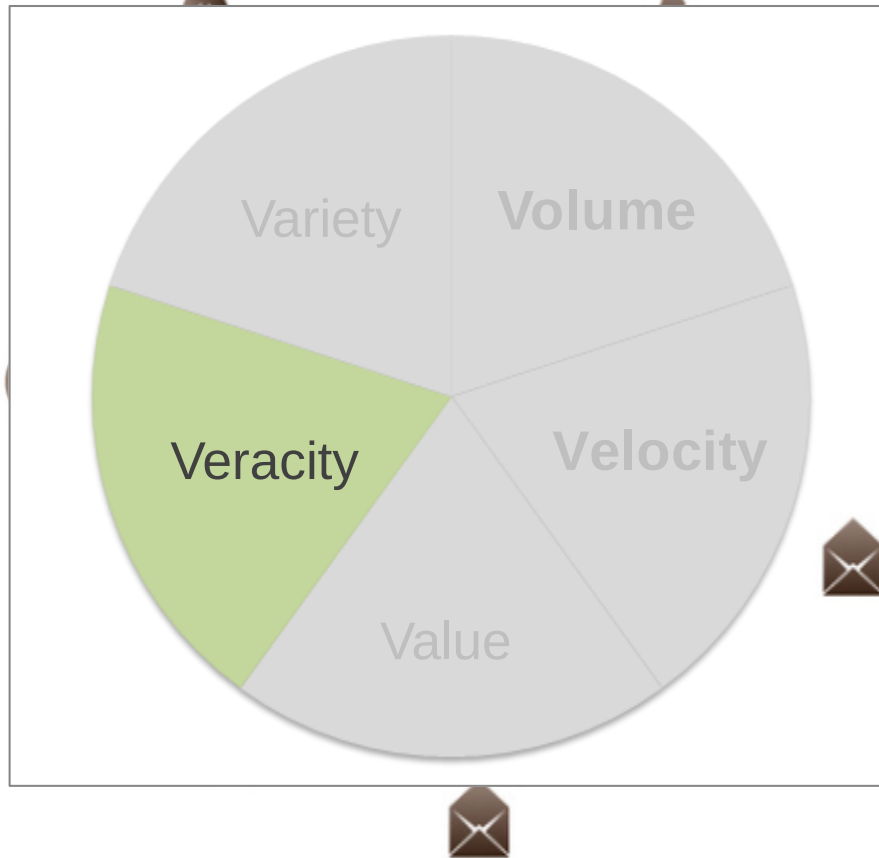
Late decisions → missing opportunities

3. Variety



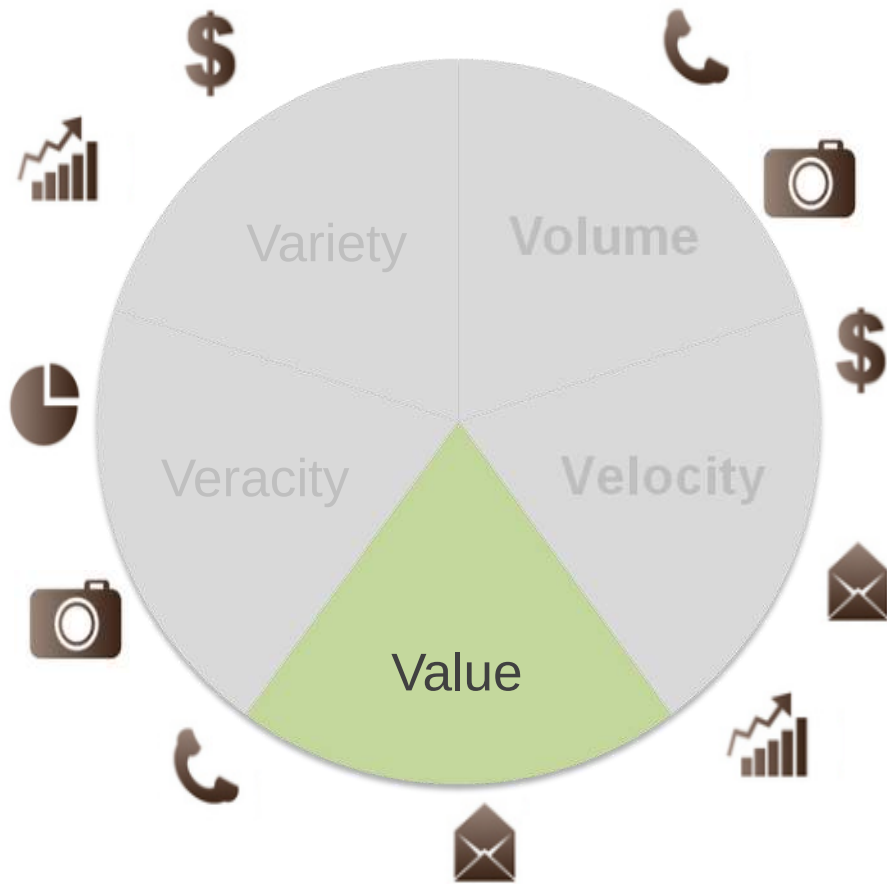
- **Variety** refers to the different types of data we can now use. Unstructured, Structured data.

4. Veracity



- **Veracity** refers to the reliability of the data generation source.
- i.e. where the data has been collected from, how it was collected

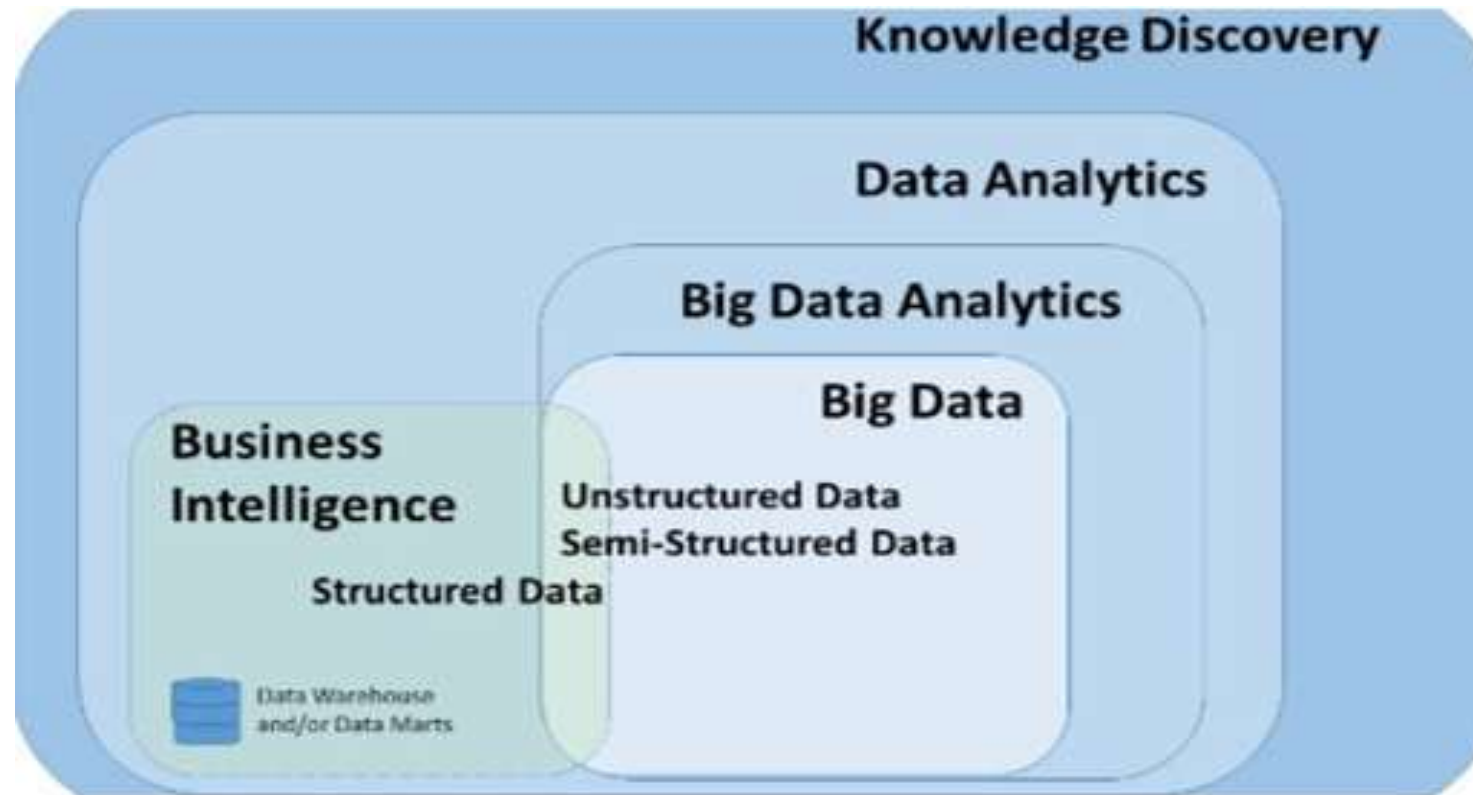
5. Value



- **Value:** Then there is another V to take into account when looking at Big Data: Value!
- Having access to big data is no good unless we can turn it into value.
- Companies have started to generate analysing value from their big data.

Big Data Analytics

- Big data analytics is the often-complex process of examining big data to uncover information -- such as hidden patterns, correlations, market trends and customer preferences -- that can help organizations make informed business decisions.



Why You Need Big Data Analytics for Your Business!

- Empowers management to make better decisions.
- Helps to identify trends and stay competitive in the market.
- Increase the business values of the organization.
- Identifies root causes of failures and then act upon them.
- Helps in predicting the future and then formulating new business policies.

Real life usage of Big Data Analytics

1. Healthcare

- **Personalized Medicine:** Big data analytics helps in creating personalized treatment plans by analyzing patient data, including genetics, medical history, and lifestyle factors. For example, IBM Watson Health uses big data to recommend personalized cancer treatments.

2. Retail and E-commerce

- **Customer Personalization:** Retailers like Amazon use big data analytics to provide personalized recommendations based on customers' browsing and purchase history. This enhances customer experience and increases sales.

3. Finance and Banking

- **Fraud Detection:** Financial institutions use big data analytics to detect fraudulent transactions in real-time by analyzing patterns and anomalies in transaction data. For instance, credit card companies use machine learning algorithms to flag unusual spending patterns.

Real life usage of Big Data Analytics

4. Telecommunications

- **Churn Prediction:** Telecom companies use big data to analyze customer behavior and predict which customers are likely to churn. This allows them to take proactive measures to retain customers.

5. Manufacturing

- **Predictive Maintenance:** Manufacturers use big data analytics to predict equipment failures and schedule maintenance before issues arise. This reduces downtime and maintenance costs.

6. Transportation and Logistics

- **Route Optimization:** Companies like UPS use big data to optimize delivery routes, reducing fuel consumption and improving delivery times.

7. Marketing and Advertising

- **Targeted Advertising:** Marketers use big data to analyze consumer behavior and preferences, allowing for highly targeted and personalized advertising campaigns. For example, social media platforms like Facebook use big data to serve relevant ads to users.

Big Data Architecture

INPUT/SOURCE



IOT



SENSORS



MOBILE



WWW



RBMS



MACHINE LOGS

THE BIG DATA TECHNOLOGY STACK

INGEST



PROCESSING ANALYSIS



STORAGE VOLUME



VISUALIZATION



Node.js D3



Apache Zeppelin



BANANA DASHBOARD



METABASE

Thanks Note



THANKS